



Systems Engineering (SE) Analysis & Control: Work Breakdown Structure (WBS)

TOPIC LEARNING OBJECTIVES

- Upon successful completion of this topic, the student will be able to:
- 1. Recognize a WBS as a product-oriented hierarchy and an output of the systems engineering process.
 - 2. Identify the role of the WBS in the systems engineering process.
 - 3. Recognize WBS's applicability throughout the acquisition life-cycle and across all acquisition management disciplines (e.g., technical/risk management, contracting, financial and business management, and acquisition planning).
 - 4. Recognize that MIL STD 881 provides guidance for developing a WBS.
 - 5. Recognize the WBS format.
 - 6. Identify the two types of WBS (Program and Contract).
 - 7. Recognize the relationship between the Program and Contract WBS.
 - 8. Identify who is responsible for the development and maintenance of the two types of WBS (Program and Contract).

STUDENT PREPARATION

- Student Support Material
- 1. None
- Primary References
- 1. DoD 5000 Series
 - 2. MIL STD 881 (WBS)
 - 3. MIL HDBK 245 (SOW)
- Additional References
- 1. In Class Video: Project Management: What is a Work Breakdown Structure?
<https://www.youtube.com/watch?v=wEWhnodF6ig>



Overview

- Role of the WBS
- WBS Hierarchy
- WBS Types



Introduction

- Whenever an organization has a large project to manage, breaking down the effort into manageable parts is the first step
- DoW uses a specific format, called a Work Breakdown Structure (WBS), to organize the breakdown of work into small areas and parts
- The WBS is a **valuable program management tool**
 - Used throughout all life-cycle phases
 - Manages risk by providing insight into technical aspects of program management
 - Benefits all acquisition disciplines (e.g., program management, contracting, life-cycle logistics, finance, budgeting)



Basic Role of WBS

- Both DoW and Contractors use WBS to establish a foundation for:
 - Developing program and technical plans through the Systems Engineering process
 - Developing acquisition strategy and contracting documents
 - Establishing schedules
 - Estimating costs and formulating budgets
 - Planning logistics
 - Tracking progress and accomplishments
 - Reporting progress status and analyzing problems
 - Establishing Program Management Baseline (PMB) for Earned Value Management (EVM)



Basic Role of WBS

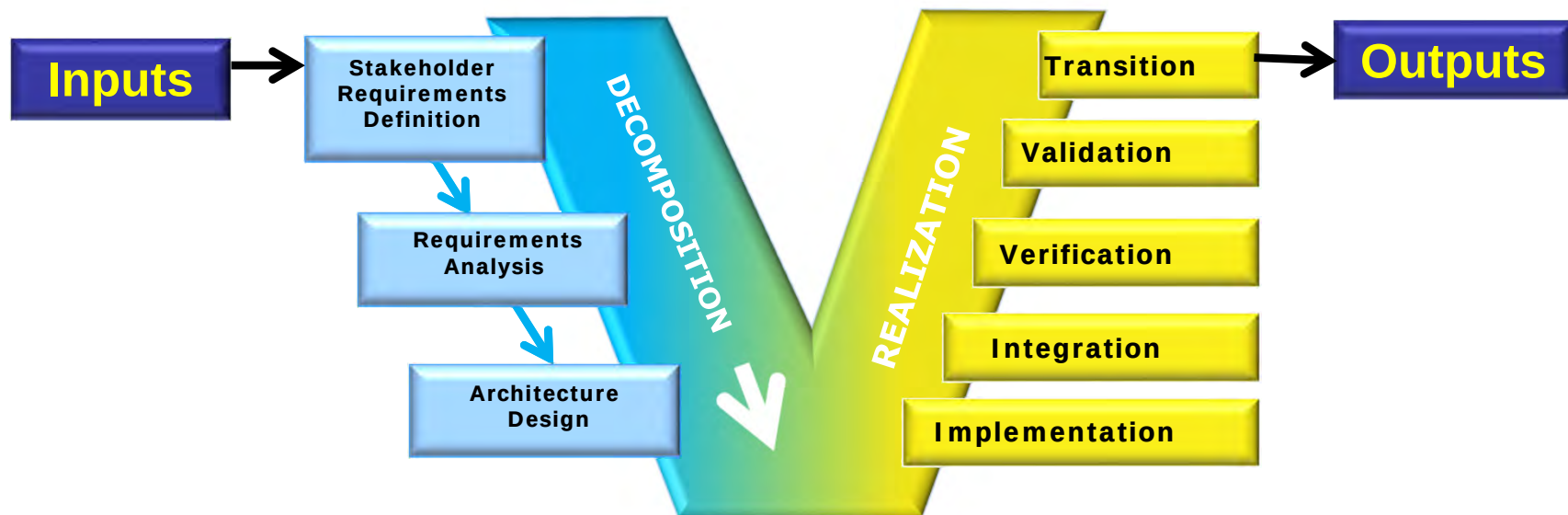
- Program Managers (PMs) use the WBS as a roadmap for Integrated Product and Process Development of the program
- PMs use the WBS to help stay within budget by identifying and analyzing tradeoffs through monitoring things such as:
 - Cost
 - Schedule
 - Performance
 - Manufacturing
 - Life-cycle Product Support
 - Testing
 - Risk management

WBS is a management tool used throughout the life-cycle and across all management disciplines



WBS and the SE Process

- WBS is an output of the SE Process



- SE Process inputs
 - Customer needs/objectives/requirements
 - Technology base
 - Output requirements from prior application of SE Process
 - Program decision requirements
 - Commercial standards & performance specs
- SE Process outputs
 - Decision data base
 - Work Breakdown Structure (WBS)**
 - System/configuration item architectures
 - Program-unique specifications & configuration baselines



WBS and the SE Process (cont.)

- The WBS supports the SE process in multiple ways to include, but not limited to:
 - Organizing IPT structure
 - Grouping system specifications
 - Structuring technical reviews
 - Understanding the impact of Engineering Change Proposals (ECPs)
 - Managing interface controls throughout the system
 - Conducting risk management, including providing additional attention to candidate risks that occur at relatively high levels and those that affect integration



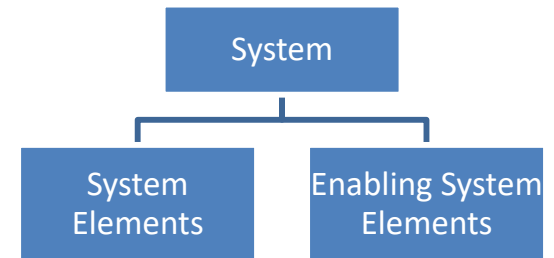
Overview

- Role of the WBS
- WBS Hierarchy
- WBS Types



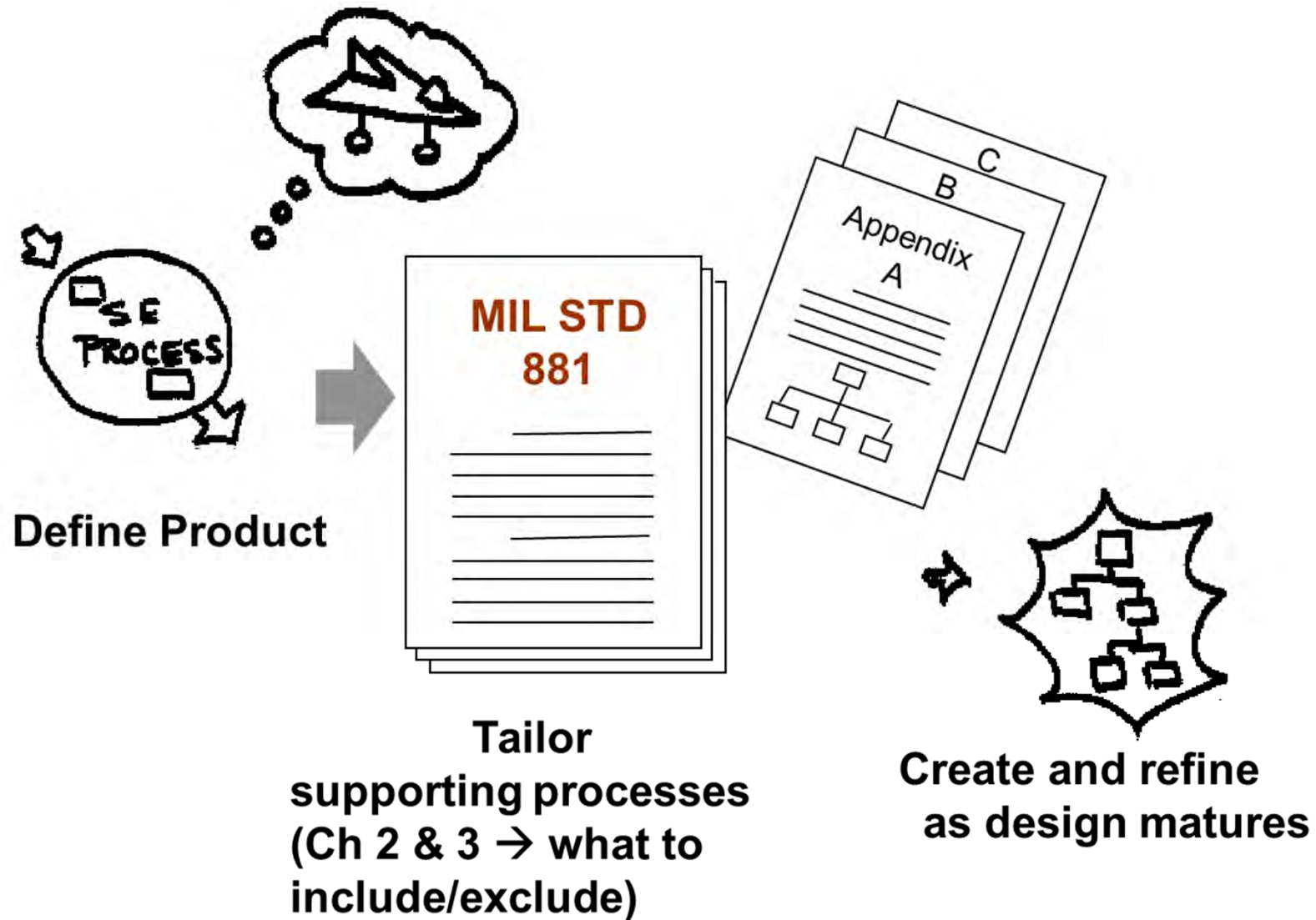
Work Breakdown Structure Hierarchy

- WBS is displayed as **hierarchically related, product-oriented** elements and the work processes required for their completion
 - Displays and defines the product or service to be developed, produced or provided
 - Relates the elements of work to be accomplished to each other and to the end product
 - Developed for each program and each individual contract within the program
 - Organizes system development activities based on the system decompositions
- A system is an aggregation of:
 - System elements
 - Enabling system elements
- **System element** refers to the **product parts** of the WBS
 - Including subsystems, components, assemblies, parts
 - Represents how the system is decomposed into Configuration Items (CIs)
 - Vertical aspect of the WBS
- **Enabling system elements** refer to common elements that provide the means for realizing the product:
 - Includes processes, equipment, parts, and facilities
 - Common elements are common to all acquisition programs developed by the DoW
 - Enable realization of the system
 - Horizontal aspect of the WBS





How to Create a WBS



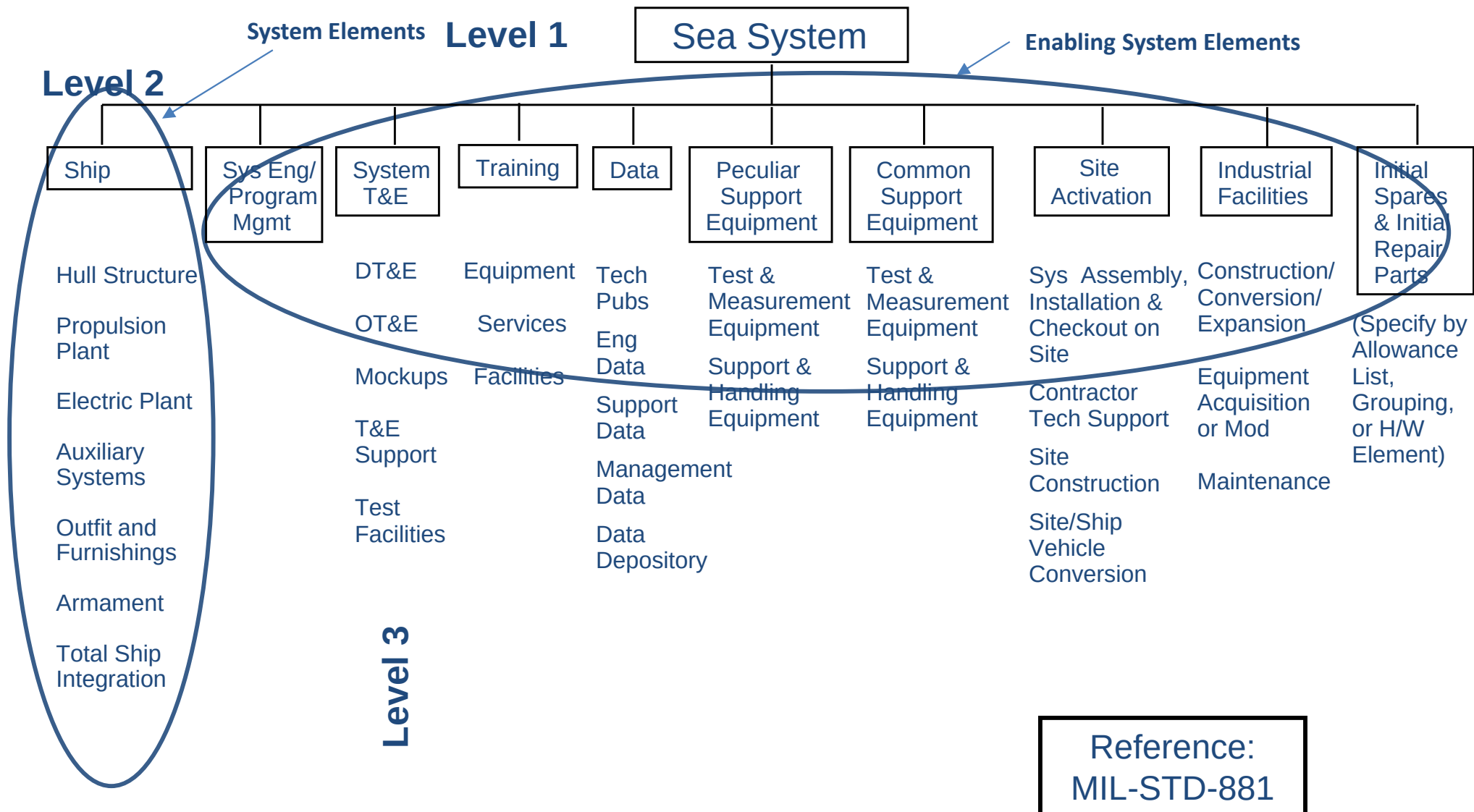


WBS Guidance

- DoW provides guidance for the development of a **standardized WBS**
- The principal document is **MIL-STD-881, DoD Standard Practice Work Breakdown Structures for Defense Materiel Items**
 - Provides uniform and consistent approach to a program's data structure
 - Assists with communication between the Government and the Contractor throughout the life-cycle of the program
- Format provided for 11 types of systems WBS
 - Aircraft Systems
 - Electronic Systems
 - Missile Systems
 - Ordnance Systems
 - Sea Systems
 - Space Systems
 - Surface Vehicle Systems
 - Unmanned Air Vehicle Systems
 - Unmanned Maritime Systems
 - Launch Vehicle Systems
 - Automated Information Systems

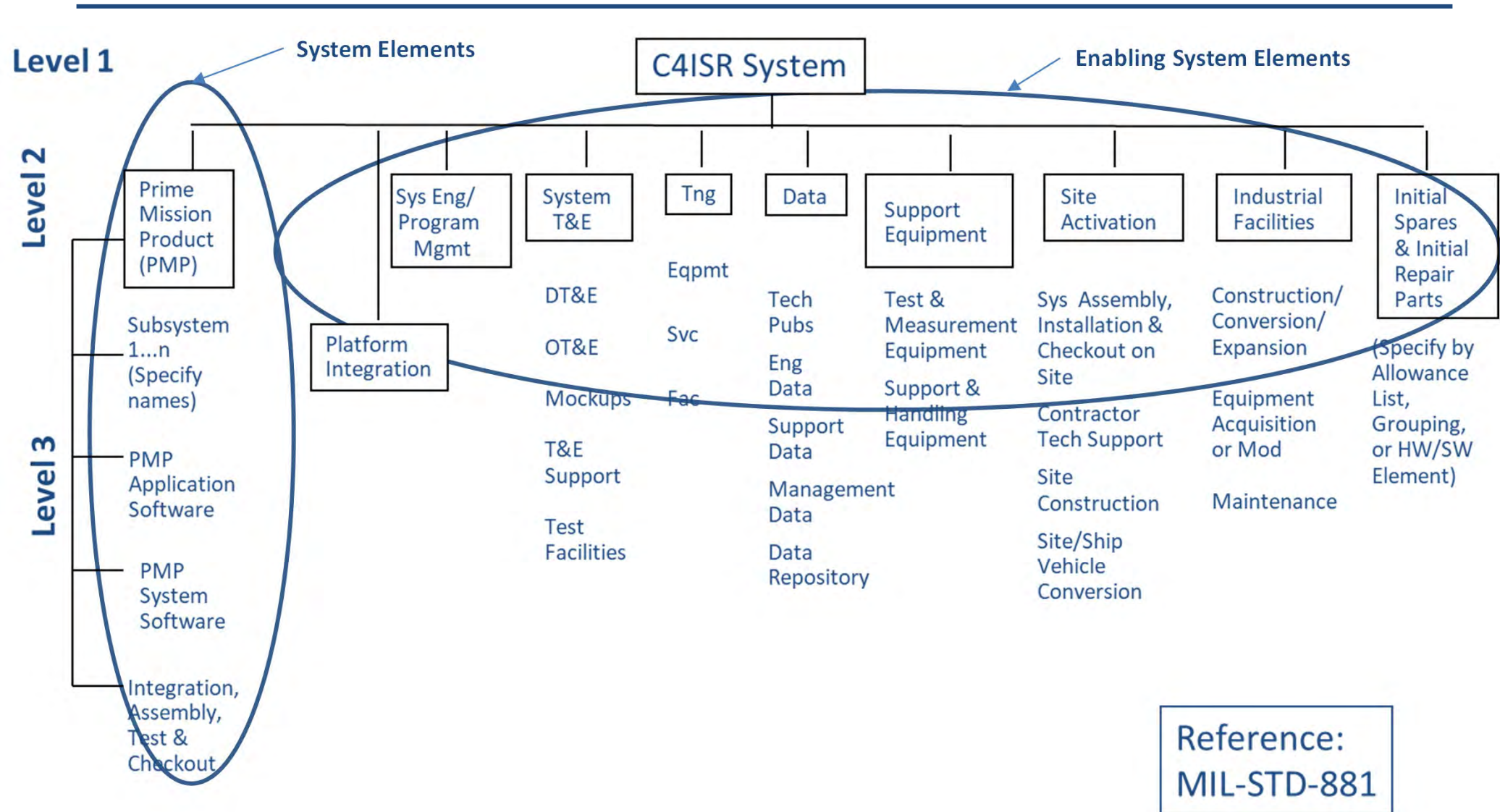


Example: Sea System WBS





Example: C4ISR System WBS





Overview

- Role of the WBS
- WBS Hierarchy
- WBS Types



WBS Types

- Types of WBS are categorized by who controls and maintains the WBS
- Two types:
 - **Program WBS** is controlled and maintained by the Program Management Office (PMO)
 - **Contract WBS** is controlled and maintained by the Contractor



Program WBS

- Program WBS encompasses the entire program, including Contract WBS and other Government elements (Operations, Manpower, Government Furnished Equipment, and Testing)
- Prepared and maintained by the Program Office
 - Derived from MIL-STD-881
 - Tailored to each specific program
 - Provided as a basis for developing the Contract WBS
- Provides a framework for specifying program objectives
 - Defines total program
 - Basis for measuring technical progress, planning technical reviews, and assessing cost and schedule performance
- Consists of levels 1-3

PM is responsible for the Program WBS



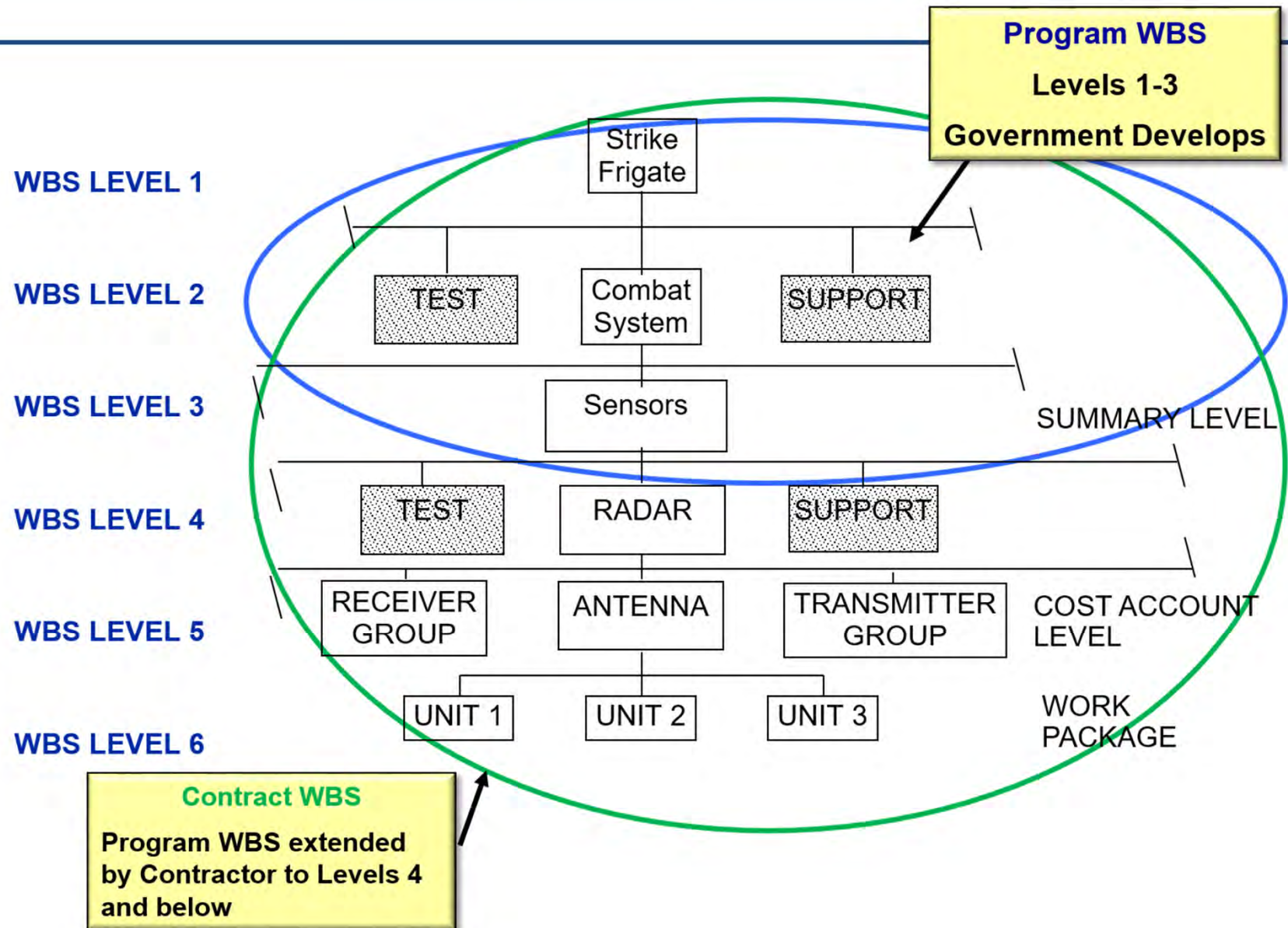
Contract WBS

- Contract WBS defines that part of the program that is being produced by a given Contractor and is the basis for collecting cost and schedule data for the contract
 - Contractor extends the Program WBS to a lower level
 - Provides management and cost information to the Government
 - Includes all the elements for products (e.g., hardware, software, data, or services) that are the responsibility of the Contractor
 - Must be consistent with the Program WBS
 - Is prepared by the Prime Contractor and should also include all Subcontractor inputs
- Contractors may extend the work breakdown structure to whatever level is necessary to manage the program

Contract WBS begins with Level 4 and can be broken down to further levels if needed

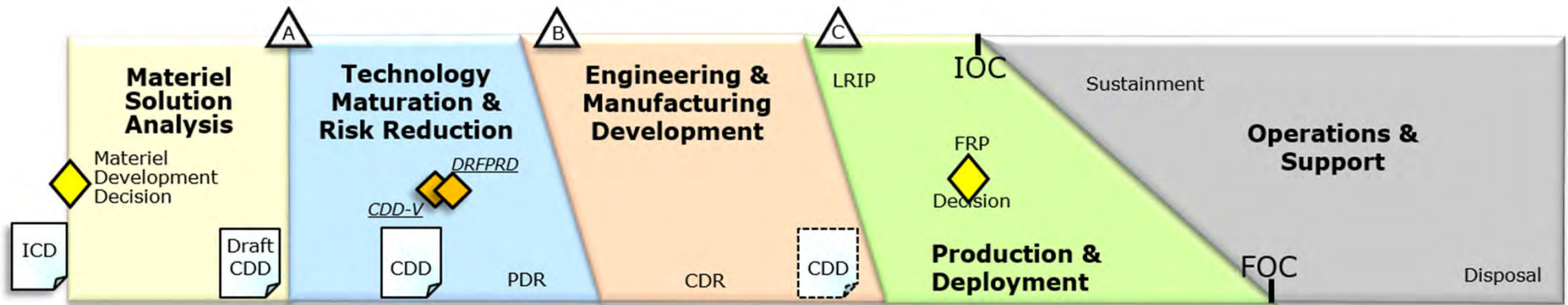


Two Types of WBS

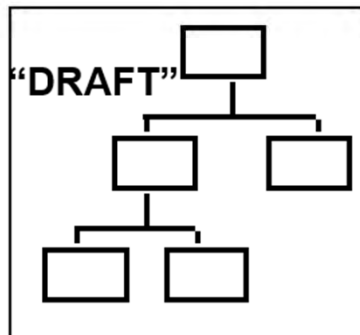




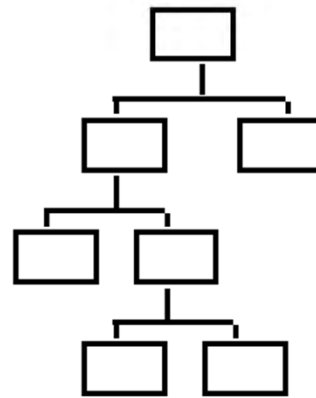
WBS Evolution Across the Life Cycle



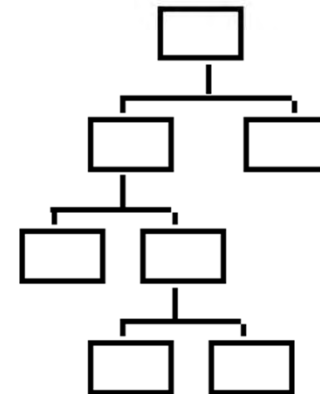
- GOVT proposes WBS
- KTR studies & submits DRAFT WBS



- KTR updates WBS
- GOVT approves updated WBS



- Final design goes into production
- GOVT approves WBS for production





Summary

- WBS is a _____ hierarchy and an _____ of the systems engineering process.
- PMs use the WBS to stay within budget by monitoring:
- Guidance for developing a WBS can be found in
- What are the 2 types of WBS?
- What is the relationship between Program & Contract WBS?
- Who is responsible for the development and maintenance of the two types of WBS?